

ANDREAS PANAYI

Athens, Greece | Limassol, Cyprus

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EDUCATION

National and Kapodistrian University of Athens (NKUA) 2024 – 2026

National Technical University of Athens (NTUA)

M.S. in Theoretical Computer Science (ALMA) GPA: 9.89/10

Selected coursework: Advanced Algorithms & Data Structures, Computational Complexity,
Graph Theory, Machine Learning, Artificial Intelligence, Algorithmic Data Science

National and Kapodistrian University of Athens (NKUA) 2020 – 2024

B.Sc. in Mathematics GPA: 8.84/10

EXPERIENCE

Research and Innovation Foundation of Cyprus, Rinnoco Ltd Jun 2026–Jul 2026

Research and Development, Systems Engineer Intern

- Research and proof-of-concept implementation of a database compression system, working at the intersection of Big Data and Machine Learning systems development.
- Built the full system in C++20 with MLPack, PostgreSQL, and Docker, using Python for testing, and applying statistical approaches alongside supervised and unsupervised machine learning techniques.

Archimedes Research Unit, Athena Research Center Jun 2025 – Jan 2026

Intern Researcher

- Designed and implemented advanced data structures for computational geometry problems in a research setting, with applications to multi-objective optimization.
- Contributed to algorithmic work across algorithms, artificial intelligence, and data science through technical analysis and implementation.

National and Kapodistrian University of Athens 2021–2026

Teaching Assistant, Computer Science I

- Conducted undergraduate laboratory sessions for the mandatory course *Computer Science I* in the Department of Mathematics.
- Taught programming in MATLAB (2021–2024) and Python (2024–2026), as well as $\text{\LaTeX}$, computer arithmetic, and numerical linear algebra.

PROJECTS

Marabu Blockchain Implementation Completed

TypeScript, C++, Linux, SSH, GitHub

- Implemented a complete blockchain node based on the Marabu protocol in TypeScript and C++.
- Deployed, tested, and debugged the system on a remote Vultr server using SSH and GitHub-based workflows.

Personal Webpage & Coursework Archive Ongoing

GitHub Pages, HTML, CSS, XeLaTeX

- Built and maintain a personal portfolio website showcasing research work, publications, and academic profile, using HTML and CSS.

- Built and maintain an online archive of coursework, university projects, and exercises; authored introductory $\text{\LaTeX}$ and $\text{\XeTeX}$ notes taught in university courses.

RESEARCH

Listing Even Cycles Beyond the Submodular Width Barrier Submitted: PODS 2027
Manuscript; joint work with Prof. Vasilis Nakos and Dr. Hung Q. Ngo

- Developed output-sensitive algorithms for listing even-cycle queries, connecting graph algorithms with database-theoretic enumeration beyond the submodular width barrier.
- Submitted to SIGMOD/PODS 2027, the flagship conference for data management.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, C++
Working Knowledge: MATLAB, Julia, R, Haskell, Prolog
Tools: Git, GitHub, SSH, Linux, Docker, $\text{\LaTeX}$, $\text{\XeTeX}$
Languages: Greek (native), English (fluent)

SCHOLARSHIPS & ATHLETICS

Cyprus State Scholarships Foundation 2020 – 2026
 Full scholarship for the entire duration of bachelor's and master's studies

Hellenic Bank Cyprus 2020 – 2024
 Full scholarship for undergraduate studies

Athletics Distinctions
 Represented Cyprus in international competitions across all age categories and earned distinctions in individual and relay events. Pancyprian champion and gold medalist in 100m, 300m hurdles, 400m, and 400m hurdles in school and club competitions.

ADDITIONAL

Military Service Jul 2019 – Sep 2020
 Served in the technical department during 14 months of military service in Cyprus